

Smart Socket:

Smart Power Distribution System For Food Trucks Using RFID-Enabled Smart Sockets.

1. Project Objective

To develop a robust software ecosystem (mobile + web) for managing user access, billing, consumption, and issue tracking across multiple smart power sockets installed in urban infrastructure (e.g., streetlight poles), primarily serving food trucks.

2. Customer Mobile App

A mobile app for **food truck operators** to monitor their electricity usage, billing, pricing, and report issues.

Map View location of all Charging spots (Live Status)

Key Features:

Feature	Description
User Login	Login via phone number + OTP or email/password
Live Session View	See current session status (power usage, cost, socket ID, region)
Usage History	List of past sessions with time, kWh used, cost
Tariff Viewer	Show pricing based on the current or selected region
User Profile	View name, contact, RFID tag, and account info

Report an Issue	Submit hardware/billing-related issues with socket/session details and image uploads
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Support Section	Help, FAQ, contact details
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Announcement/Admin-User Updates.

- Fetch announcements from backend
- Display in “Notifications” or “Messages” section
- Unread indicators (optional)
- Push notifications when a new announcement is sent

3. Admin Web Portal

A browser-based dashboard for **system administrators** to manage users, sockets, sessions, pricing, and reported issues.

Admin Roles & Access Levels

Role	Access Scope
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Super Admin	Full access to all features across all regions
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Region Admin	Manages one region’s sockets, users, pricing, and issues
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Block Admin	Manages sockets and users in assigned blocks (sub-regions)
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Admin Features by Module

Region & Block Management

- Assign admins to regions or blocks
- Define socket deployment zones

User Management

- View all users by region/block
- Activate/deactivate user accounts
- View user-specific reports

Session Management

- View historical and live charging sessions
- Filter by region, user, socket, or date
- Export sessions to Excel or CSV

Region-Wise Pricing

- Define per-kWh pricing per region or block
- Add time-based pricing schedules (peak/off-peak)
- View and update pricing history

Reports & Analytics

- Aggregate reports by:
 - Region / Block / Socket / User
 - Daily / Weekly / Monthly / Custom ranges

- Exportable formats (PDF, Excel)
- Usage charts, revenue breakdowns

Issue Tracking System

- View issues submitted by users
- Filter by region, user, type, or status
- Assign issues to admins for resolution
- Change issue status (Open > In Progress > Resolved)
- Add resolution notes
- Upload documentation or photos

Role & Access Control

- Create or manage admin users
- Assign admin roles and region/block scopes

Create Announcement

Field	Input Type
Title	Text input
Message Body	Rich text editor
Target Audience	All / Region / Block / Specific users
Schedule Time	(Optional) Future scheduling

Attachments

Optional image or PDF (e.g., tariff notice)

Manage Announcements

- View all past announcements
 - Filter by region/date/user
 - Edit or delete (if still scheduled)
 - View delivery status (e.g., sent/pending)
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4. System Components & Architecture

Backend Services

- Built using **Node.js**+express
- REST APIs for mobile + web
- JWT-based auth for admins & users
- Role-based access permissions
- Socket-device communication via separate channel (not part of this software)

Database

- **PostgreSQL** or **MongoDB**
- Key tables: Users, Regions, Blocks, Sockets, Sessions, Issues, Admins, Pricing

Cloud Hosting & DevOps

- AWS, DigitalOcean, or GCP deployment
 - S3 or Firebase Storage for image uploads (issue photos)
 - Daily/weekly backups
 - MapBox/Google Maps for Socket map view
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5. Security & Compliance

- All APIs protected by token-based auth
 - Role-based data filtering for admin views
 - Optional audit logs of admin actions
 - Secure image/document upload
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Project Phasing Plan

Phase 1: Static MVP (Presentation-Ready App)

Goal: **Demonstrate UI, flow, and concept** using dummy/static data.
Customer Mobile App (Static)

Feature	Details
Login	Static UI (no actual auth)
Dashboard	Show fake session in progress (dummy kWh, timer, region)

Usage History Dummy list of 3–5 past sessions

Pricing Viewer Static region dropdown with mock per-kWh prices

Profile Static user info + RFID ID

Report Issue Opens form, shows mock confirmation

UI Polish Clean, modern React Native UI for demo

Admin Portal (Static)

Feature	Details
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Login Page	Dummy login (skip logic)
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Dashboard	Static overview: Users = 20, Active Sockets = 12, kWh used = 1024
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User List	10 mock users with RFID + region
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Region Pricing	Show 3 sample regions with editable price fields
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Sessions	Static table with fake session entries
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Issue Tracker 2 dummy issue cards with dropdown status changer

Role UI Show Admin types and dummy region/block assignments

Phase 2: Full Logic & Integration Development

Goal: **Production-ready system** with real-time data, backend APIs, admin roles, and region logic.

Mobile App Features

- Real login/authentication (Firebase or backend JWT)
- Live session tracking (from backend/smart socket)
- Dynamic region-wise pricing
- Real-time usage history from API
- Report issue → stored in DB + admin alerts

Admin Portal Features

- Real user/session/socket data
- Role-based admin access
- Create/edit regions and assign admins
- Dynamic pricing per region/block
- Issue resolution workflow
- Real-time session table with filters and search

- Export reports

Backend Enhancements

- Full API layer with auth, CRUD, and filters
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 - Admin auth + role-based API scoping
 - Relational DB schema (PostgreSQL preferred)
 - API rate limits and security
 - Email + push notification integration
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